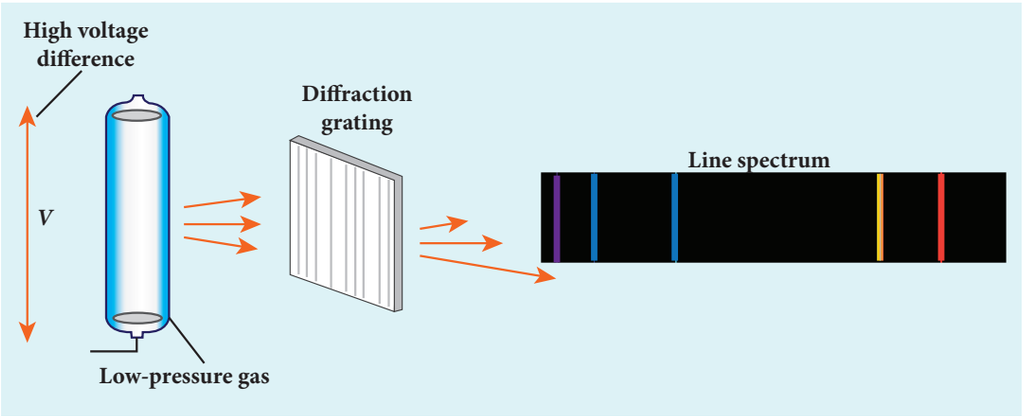


Figure 9.21 Spectrum of an atom

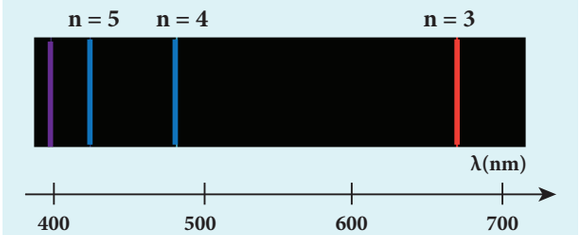
electrical discharge. When they were viewed or photographed, instead of a continuous spectrum, the radiation contains of a set of discrete lines, each with characteristic wavelength. In other words, the wavelengths of the radiation obtained are well defined and their positions and intensities are characteristic of the element as shown in Figure 9.21.

This implies that these spectra are unique to each element and can be used to identify the element of the gas (like finger print used to identify a person) – that is, it varies from one gas to another gas. This uniqueness of line spectra of elements made the scientists to determine the composition of stars, sun and also used to identify the unknown compounds.

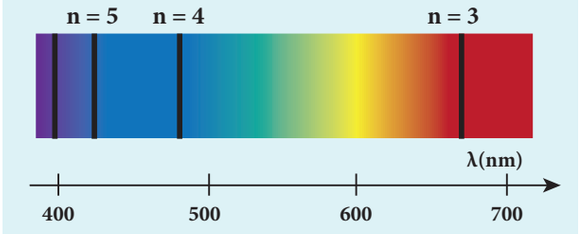
Hydrogen spectrum

When the hydrogen gas enclosed in a tube is heated up, it emits electromagnetic radiations of certain sharply-defined characteristic wavelength (line spectrum), called hydrogen emission spectrum (Refer unit 5, volume 1 of +2 physics text book). The emission spectrum of hydrogen is shown in Figure 9.22(a).

When any gas is heated up, the thermal energy is supplied to excite the electrons. Similarly by all occurring light on the atoms, electrons can be excited. Once the electrons get sufficient energy as given by Bohr's postulate (c), it absorbs energy with particular wavelength (or frequency) and jumps from one stationary state (original state) to another state with those wavelengths (or frequencies) for the colours that are not observed are seen as dark lines in the absorption spectrum as shown in Figure 9.22 (b).



(a) Emission spectrum of hydrogen (in wavelength)



(b) Absorption spectrum of hydrogen (in wavelength)

Figure 9.22 Hydrogen spectrum (a) emission (b) absorption

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